Pre S3 L07-09 Area and Volume Quiz

(ANSWER)

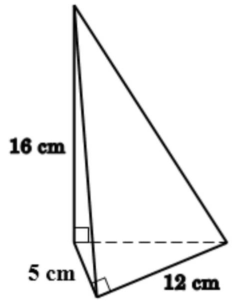
/34

Unit Education

Name: ____________________

Date: ____________________

1. What is the volume of pyramid in the figure?



A. 320 cm^3

B. 230 cm^3

C. 160 cm^3

D. 80 cm^3

Volume of the pyramid $= \frac{1}{3} \times \left(\frac{1}{2} \times 5 \times 12\right) \times 16\text{ cm}^3 = \underline{\underline{160\text{ cm}^3}}$

C

2. In the figure, the height of the right circular cone is 4 cm and its base radius is 3 cm. The base radius of the cylinder is 2 cm. If they have the same volume, find the height of the cylinder.

A. 2 cm

B. 3 cm

C. 4 cm

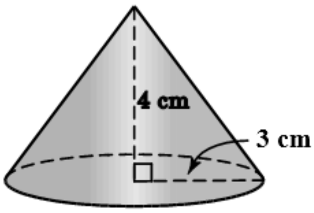
D. 6 cm

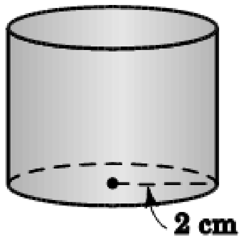
Let h cm be the height of the cylinder.

$$\pi \times 2^2 \times h = \frac{1}{3} \times \pi \times 3^2 \times 4$$

$$h = 3$$

$\therefore$ The height of the cylinder is 3 cm.





B

3. 27 small solid metal spheres each of radius 3 cm are melted and recast into a larger solid metal sphere. Find the surface area of the large sphere in terms of π.

A. $36\pi\text{ cm}^2$

B. $162\pi\text{ cm}^2$

C. $324\pi\text{ cm}^2$

D. $972\pi\text{ cm}^2$

Let r cm be the radius of the large sphere.

Volume of the large sphere $= \frac{4}{3}\pi r^3$

$$27 \times \frac{4}{3} \times \pi \times 3^3 = \frac{4}{3}\pi r^3$$

$$r^3 = 729$$

$$r = 9$$

Surface area of the large sphere

$= 4 \times \pi \times 9^2\text{ cm}^2$

$= \underline{\underline{324\pi\text{ cm}^2}}$

C

4. The surface area of a sphere is $64\pi\text{ cm}^2$. Find the volume of the sphere.

A. $\frac{64}{3}\pi\text{ cm}^3$

B. $\frac{128}{3}\pi\text{ cm}^3$

C. $64\pi\text{ cm}^3$

D. $\frac{256}{3}\pi\text{ cm}^3$

Let r cm be the radius of the sphere.

$4\pi r^2 = 64\pi$

$r^2 = 16$

$r = 4$

$\therefore$ The radius of the sphere is 4 cm.

Volume of the sphere $= \frac{4}{3} \times \pi \times 4^3\text{ cm}^3 = \underline{\underline{\frac{256}{3}\pi\text{ cm}^3}}$

D

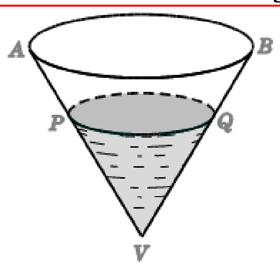
5. A vessel in the shape of an inverted right circular cone containing 324 cm^3 of water is placed vertically. If the depth of water in the vessel is $\frac{3}{5}$ of the height of the vessel, find the capacity of the vessel.

A. 540 cm^3

B. 750 cm^3

C. 1250 cm^3

D. 1500 cm^3



Let $x\text{ cm}^3$ be the capacity of the vessel.

$\because$ Cones VPQ and VAB are similar.

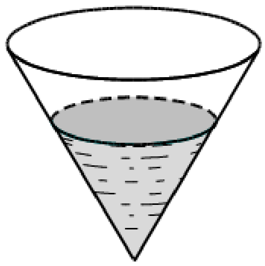
$\therefore$ $\frac{324\text{ cm}^3}{x\text{ cm}^3} = \left(\frac{3}{5}\right)^3$

$$\frac{324}{x} = \frac{27}{125}$$

$$x = 324 \times \frac{125}{27}$$

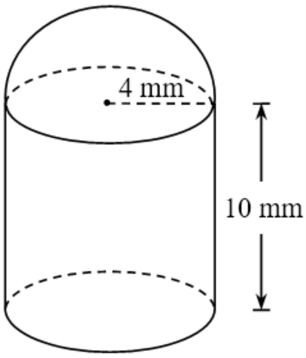
$$= 1\ 500$$

$\therefore$ The capacity of the vessel is $1\ 500\text{ cm}^3$.



D

6. The figure shows a small component of a plastic model. The shapes of its upper part and lower part are a hemisphere and a cylinder respectively. Their base radii are 4 mm and the height of the cylinder is 10 mm. Find the volume of the component.
(Give the answer correct to 3 significant figures.) (3 marks)



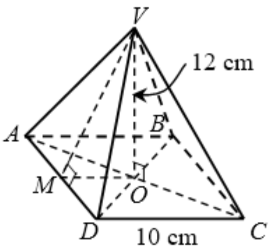
Volume of the component

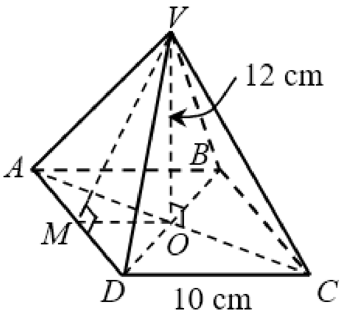
= volume of the hemisphere + volume of the cylinder

$= \left(\frac{1}{2}\times\frac{4}{3}\times\pi\times4^3+\pi\times4^2\times10\right)$ mm^3 1M+1M

= $\underline{\underline{637 \text{ mm}^3}}$, *cor. to 3 sig. fig* 1A

7. The height of a solid right pyramid with a square base of side 10 cm is 12 cm. Find the total surface area of the pyramid. (4 marks)





Refer to the notations in the figure.

$OM = \frac{1}{2}\times10$ cm = 5 cm 1M

In $\triangle VOM$,

$VM = \sqrt{MO^2+VO^2}$ (*Pyth. theorem*) 1M

$= \sqrt{5^2+12^2}$ cm

= 13 cm

Total surface area of the pyramid

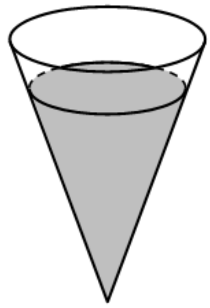
= 4 × area of $\triangle VAD$ + area of square $ABCD$

$= 4\times\frac{1}{2}\times AD\times VM+CD^2$

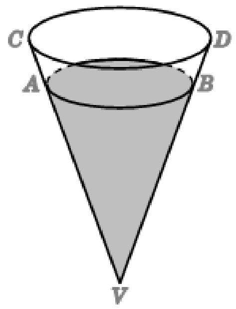
$= \left(4\times\frac{1}{2}\times10\times13+10^2\right)$ cm^2 1M

= $\underline{\underline{360 \text{ cm}^2}}$ 1A

8. The figure shows a vertical right conical vessel containing some water. The ratio of the curved surface area of the vessel to area of the surface of the vessel in contact with water is $36:25$.
   (a) Find the ratio of the base radius of the vessel to the radius of the water surface. (2 marks)
   (b) It is given that the vessel contains 600 cm^3 of water. Alex claims that if 300 cm^3 of water is added into the vessel, the water will overflow. Do you agree? Explain your answer. (5 marks)



Refer to the notations in the figure.



(a) Let x and y be the base radius of the vessel and the radius of the water surface respectively.

$\because$ Cones VAB and VCD are similar.

$$\therefore \left(\frac{x}{y}\right)^2 = \frac{36}{25} \quad \text{1M}$$

$$\frac{x}{y} = \frac{6}{5}$$

$\therefore$ <u>The required ratio is 6 : 5.</u> 1A

(b) Let $z\text{ cm}^3$ be the capacity of the vessel.

$$\frac{z}{600} = \left(\frac{6}{5}\right)^3 \quad \text{1M}$$

$$\frac{z}{600} = \frac{216}{125}$$

$$z = 1\,036.8 \quad \text{1A}$$

$\therefore$ Volume of the remaining space in the vessel $= (1\,036.8 - 600)\text{ cm}^3$

$= 436.8\text{ cm}^3$ 1A

$> 300\text{ cm}^3$ 1M

$\therefore$ Water will not overflow.

$\therefore$ <u>The claim is disagreed.</u> 1A